

<[Zr₆(μ₃-O)₄(μ₃-OH)₄(O₂C-C₆H₄-CO₂)₁₂], UiO-66

Materials

50 mL round bottom flask
Two centrifuge tubes

Chemicals

Zirconium(IV) chloride ZrCl₄
Acetic Acid CH₃COOH
Hydrochloric acid 37%
Benzoic Acid
Dimethylformamide (p.a.)
Terephthalic acid (H₂BDC) C₈H₆O₄
Ethanol (p.a.)

Synthesis

In a 50 mL round bottom flask, ZrCl₄ (0.235 g, 1.0 mmol) and either CH₃COOH solution (2.5 mL, 24 mmol) or HCl solution (2 mL, 24 mmol) or benzoic acid (2.93 g, 24 mmol) are dissolved in DMF (30 mL). H₂BDC is added to the clear solution (0.235 g, 1.4 mmol). The system is heated for 24 h at 120 °C.

After 24 h, the solution is cooled to room temperature and the precipitate is isolated by centrifugation. The solid is suspended in DMF (30 mL) and then recovered by centrifugation. This process is repeated twice. The obtained powder is washed with ethanol (2·30 mL) in the same way as described before (DMF washing step). Finally, the solids are dried under reduced pressure.

Analytatics

- Surface area and porosity measurement using the NOVA sorption analyzer. The sample should be degassed at 120 °C for 2 h.
- IR (Infrared) spectrum. The analysis should be performed in a neat sample using a KBr disk.
- PXRD (Powder X-Ray Diffraction) phase analysis
- TGA (Thermo Gravimetric Analysis). The analysis should be performed on a fresh, non degassed sample.

Control questions

- What is the structure of UiO-66? What is the coordination environment of the zirconium atom? What is the secondary building block (SBU). What does 'isorecticular' means and are there 'isorecticular' MOFs regarding to UiO-66 ?
- The compound you obtained is 'highly defected' (the number of terephthalates is close to four instead of six, which is typical for the case, when no modulator is used). What kind of structural defects do you know? How can they be characterized?

Literature

1. Katz, M. J.; Brown, Z. J.; Colón, Y. J.; Siu, P. W.; Scheidt, K. a; Snurr, R. Q.; Hupp, J. T.; Farha, O. K. A Facile Synthesis of UiO-66, UiO-67 and Their Derivatives. *Chem. Commun.* 2013, 49 (82), 9449–9451.
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3. J. H. Cavka, S. Jakobsen, U. Olsbye, N. Guillou, C. Lamberti, S. Bordiga, K. P. Lillerud, *J. Am. Chem. Soc.* 2008, 130, 13850–13851.
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